

Shortlist & Research Ideas — Multi-Agent Root-Cause Attribution

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The core 12: read these first

Together, these 12 papers cover your whole problem: defining attribution, how a compromise spreads, how to detect it, and how to trace it back. Every one is peer-reviewed at a top venue, and the authors come from groups such as UC Berkeley, CMU, Cornell, Penn State, NUS, Tsinghua, Google DeepMind and Microsoft. I checked each venue on the conference's own page, and each affiliation on the paper itself. **If you only have time for five, read 1, 2, 3, 4 and 11.**

#	Paper	Venue	Institutions	Why read it
1	Which Agent Causes Task Failures and When? (Who&When)	ICML 2025, Spotlight	Penn State, Duke, Google DeepMind, Meta, Univ. of Washington	Defines "which agent, which step" and releases the dataset everyone benchmarks against.
2	Why Do Multi-Agent LLM Systems Fail? (MAST)	NeurIPS 2025 Datasets & Benchmarks, Spotlight	UC Berkeley (Stoica, Zaharia, Gonzalez, Klein)	A taxonomy of 14 failure modes, the vocabulary reviewers will expect you to use.
3	On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents	ICML 2025	CUHK, CMU, Tsinghua	Makes one agent faulty in chain, flat and hierarchical structures. The closest existing experiment to yours.

#	Paper	Venue	Institutions	Why read it
4	AgenTracer	ICLR 2026	NUS, CUHK, NTU	Labels the culprit by replaying runs with one step changed and by injecting faults, then trains an 8B tracer model.
5	Aegis	ICLR 2026	Peking Univ., Tsinghua, NUS	Injects errors into chosen agents at scale: 9,533 trajectories.
6	DoVer	ICLR 2026	Microsoft, Chinese Academy of Sciences	Confirms a suspected cause by editing the run and replaying it.
7	Agent Smith	ICML 2024	Sea AI Lab, NUS, Singapore Management Univ.	One poisoned agent jailbreaks up to a million agents, with infections growing exponentially.
8	Multi-Agent Systems Execute Arbitrary Malicious Code	COLM 2025	Cornell Tech (Shmatikov)	The attack enters through one agent, and a different agent carries out the harmful action.
9	G-Safeguard	ACL 2025	USTC, NUS, UCLA	A graph neural network (GNN) flags the compromised agent and cuts its links to the others.

#	Paper	Venue	Institutions	Why read it
10	GUARDIAN	NeurIPS 2025	King's College London, Tsinghua	Tracks the collaboration over time, so it finds which round as well as which agent.
11	CaMeL: Defeating Prompt Injections by Design	IEEE SaTML 2026	Google DeepMind, Google, ETH Zurich (Tramèr)	Attaches a tag recording its source to every piece of data the agent handles, a secure-by-design approach.
12	TracLLM	USENIX Security 2025	Penn State	Traces an output back to the part of the context that caused it, the forensic step at each hop.

Supporting references by topic

Cite these in related work. You don't need to read them in full: skim the abstract and the main figure. Each is peer-reviewed at a top venue or is a well-established classic.

Orchestration frameworks

Paper	Venue	Institutions	Use it for
AutoGen	COLM 2024	Microsoft Research, Penn State, Univ. of Washington	The standard framework to build your chain on.
MetaGPT	ICLR 2024, Oral	DeepWisdom, KAUST, UC Berkeley, IDSIA	Structured handoffs between agents.
Multiagent Debate	ICML 2024	MIT, Google Brain	The robustness-versus-cost baseline.

Paper	Venue	Institutions	Use it for
AgentPrune	ICLR 2025	Tongji, Shanghai AI Lab, CUHK, UNC Chapel Hill	Evidence that cutting messages lowers cost and also blocks attack paths.
MaAS	ICML 2025, Oral	NUS, Tongji, NTU, Shanghai AI Lab, USTC	Picks a system design per query at 6–45% of the usual cost.

Self-healing and recovery

Paper	Venue	Institutions	Use it for
Reflexion	NeurIPS 2023	Northeastern, MIT, Princeton	Agents that learn from their own failures.
Large Language Models Cannot Self-Correct Reasoning Yet	ICLR 2024	Google DeepMind, UIUC	Why healing needs an independent verifier.
SCoRe	ICLR 2025, Oral	Google DeepMind	Training models to correct themselves.
Self-Healing Machine Learning	NeurIPS 2024	Univ. of Cambridge (van der Schaar)	Diagnose the cause before choosing a fix.
SagaLLM	PVLDB 2025	Stanford	Transaction-style rollback across agents.
The Vision of Autonomic Computing	IEEE Computer 2003	IBM Research	The monitor-analyze-plan-execute loop (MAPE-K) that self-healing systems are built on.

Testbeds and benchmarks

Paper	Venue	Institutions	Use it for
AgentDojo	NeurIPS 2024 Datasets & Benchmarks	ETH Zurich, Invariant Labs	Prompt-injection environment to wrap as agent A.

Paper	Venue	Institutions	Use it for
ToolEmu	ICLR 2024	Univ. of Toronto, Vector Institute, Stanford	An LLM plays the tools, so testing has no real side effects.
MultiAgentBench	ACL 2025	UIUC	Benchmark with a chain topology built in.
Agent Security Bench	ICLR 2025	Zhejiang Univ., Rutgers	27 attack and defence methods to reuse.
InjecAgent	ACL 2024 Findings	UIUC	1,054 indirect prompt-injection test cases.
A2ASecBench	ICLR 2026	Duke, Johns Hopkins, Univ. of Michigan	Attacks on the Agent2Agent protocol, including forged agent identities.
TheAgentCompany	NeurIPS 2025 Datasets & Benchmarks	CMU (Neubig)	A realistic simulated workplace.
τ-bench	ICLR 2025	Sierra (Narasimhan, Yao)	The pass ^k metric: does the agent succeed on every one of k repeated tries?

Forensics and theory

Paper	Venue	Institutions	Use it for
Backtracking Intrusions	SOSP 2003	Univ. of Michigan	The classic algorithm that walks back from a symptom to where the attack entered.
NoDoze	NDSS 2019	UIUC, NEC Labs America	Ranking which dependency path is most likely the real cause.
Responsibility and Blame	JAIR 2004	Cornell (Halpern)	A formal definition of how much blame each agent deserves.

Research ideas

Five projects, each aimed at a gap the core papers leave open. The first three form one line of work: build a benchmark, trace compromises back to their source, then heal them. The last two can be separate papers. Numbers in brackets refer to the core 12.

Idea 1: An adversarial root-cause benchmark ("Who&When for attacks")

- **Gap:** Every attribution dataset uses honest mistakes [1, 4, 5]. None contains a deliberately compromised agent whose output looks normal while its payload goes off downstream.
- **What you build:** Chains of 3 to 7 agents in AutoGen or MultiAgentBench. You inject real attacks into one chosen agent: prompt injection (payloads from Agent Security Bench or InjecAgent), poisoned tools, poisoned memory, and triggers that go off only later. For each run you record the entry agent, the entry step, the agents that relayed it, and the agent where the symptom appeared. You also add honest-error runs, so a method has to tell attacks apart from mistakes.
- **How you evaluate:** Run the Who&When baselines, AgenTracer, G-Safeguard and GUARDIAN [1, 4, 9, 10]. Report where each breaks down as chains get longer and attacks get stealthier.
- **Target venues:** NeurIPS Datasets & Benchmarks track, or ICLR.

Idea 2: Backtracking attribution over signed provenance logs

- **Gap:** LLM judges read the whole trace, and their accuracy collapses on long traces. Operating-system security solved the same problem with dependency graphs (Backtracking Intrusions, NoDoze).
- **What you build:** Middleware for the orchestrator that logs every hop: each agent's inputs, outputs, tool descriptions and memory reads, each with a signed hash. From the anomalous output at C, you walk the resulting dependency graph backwards. Candidate paths are ranked by how rare their events are, and TracLLM [12] narrows each hop to the context that mattered. You confirm the top suspect by replaying the run with A's output replaced and checking whether the symptom at C disappears [6].
- **How you evaluate:** Accuracy at naming the right agent and step on Idea 1's benchmark and on Who&When. Also report token cost, and how many replays each diagnosis needs.
- **Target venues:** USENIX Security, CCS or NDSS for the security framing. ICLR or ICML for the method.

Idea 3: A self-healing loop built for compromise

- **Gap:** Detectors cut off the bad agent and stop there [9, 10]. Self-healing work handles honest faults, not attacks.
- **What you build:** A loop based on MAPE-K (monitor, analyze, plan, execute over shared knowledge). It quarantines A by revoking its tool privileges, then brings it back with a different model, a clean memory and a fresh prompt. Only the downstream steps that A's output reached are re-run, found from the provenance log in Idea 2. Side effects that already happened are undone with compensating actions, as in SagaLLM. A must pass probe tests before it is let back into the pipeline.
- **How you evaluate:** Recovery rate, time to recover, blast radius (how many agents were affected), false-quarantine rate, and cost compared with restarting the whole pipeline.
- **Target venues:** SEAMS or ICSE for the software-engineering angle, or NeurIPS or ICLR.

Idea 4: The cost-robustness frontier of orchestration

- **Gap:** Efficiency work such as AgentPrune and MaAS optimizes for cost and accuracy. No one has mapped how much robustness you lose as you cut cost when an upstream agent is compromised.
- **What you build:** A sweep over structures: chain, hierarchical, star and debate. Each is tested with and without an inspector agent [3] at different hops. One concrete question: in a chain of k agents, where is the best place for a single inspector? Then add a robustness term to a topology designer such as G-Designer or AgentPrune.
- **How you evaluate:** Accuracy, token cost and attack spread, plotted as a trade-off curve. Include a single-model baseline (chain-of-thought with self-consistency) at the same cost, which reviewers now expect.
- **Target venues:** ICML, NeurIPS or ACL.

Idea 5: Attacking the auditor

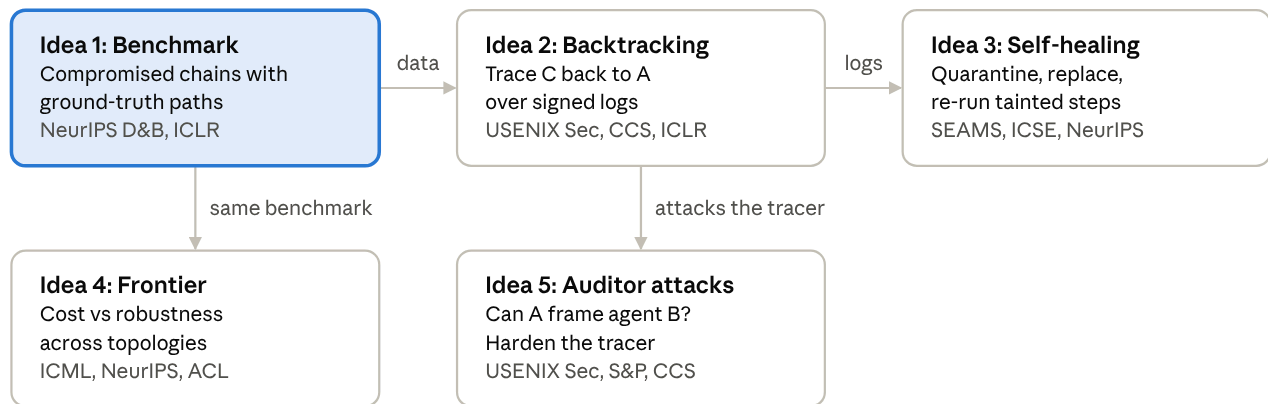
- **Gap:** No one has tested whether a compromised agent can fool the attribution system itself, for example by framing agent B, forging its own reasoning, or writing text aimed at the judge model.
- **What you build:** Adaptive attacks against LLM-judge attributors [1, 4] and graph detectors [9, 10]. Then defences: signed logs, a judge that reads untrusted content only in quarantine (the CaMeL [11] pattern), and cross-checks between the log and replays.
- **How you evaluate:** How often attribution flips to the wrong agent under attack, before and after each defence.

- **Target venues:** USENIX Security, IEEE S&P or CCS. This is the most novel idea and the best fit for security venues.

Where to start

Start with Idea 1. It is the fastest route to a publishable result, and every other idea needs its benchmark to be evaluated on.

Start with the benchmark: every other idea reuses it



how the five ideas depend on each other · with target venues

1. **First paper:** build the benchmark and reproduce Who&When and AgenTracer on it.
Your first result is showing where current methods fail on deliberate compromises.
2. **Second paper:** the backtracking attributor, evaluated on the same benchmark. Idea 5 follows naturally from it, as a security paper that attacks your own tracer.
3. **Framework:** add self-healing and the cost-robustness study to complete the orchestrator.

To make your references look strong, cite the core 12 in your introduction and threat model, and the supporting tables in related work.